職業衛生研究與趨勢議題
國際研討會

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on Occupational Health Research
and Trends

NOV.10
2026

Occupational Health Threats in Emerging Gig Economies: Potential Health Effects on Platform Workers

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辦理單位：中國勞工安全衛生管理學會

Digital Labor Platforms

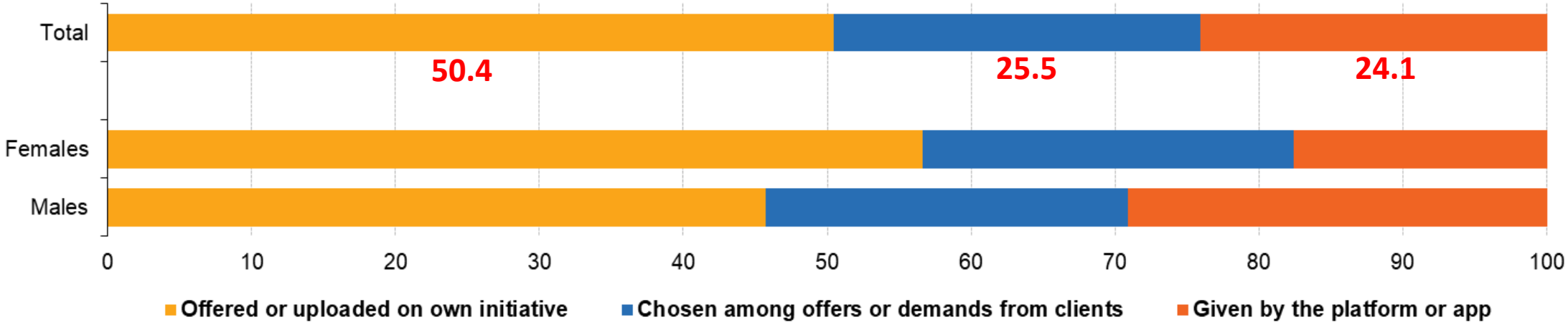
- Web-based platforms
  - Remote work
  - Programming, graphic design, or transcription
- Location-based platforms
  - Perform work at a specific location
  - Delivering of goods, taxi- or household work

Digital Labor Platforms

- A growing feature of the modern labor market
  - Resulting in new ways of organizing work
  - Changing how we understand and relate to labor
- Positive aspects
  - Lower entry barriers to work
- Relevant challenges
  - Increasing exposure to occupational health hazards among workers
  - An increasing share of precariously employed workers

The European Union Labour Force Survey (EU-LFS), 2022

Digital platform workers by type of assignment of the work, service or task
(as % of all digital platform workers for at least 1h in the last month, aged 15-64, Agg 17(¹), 2022)

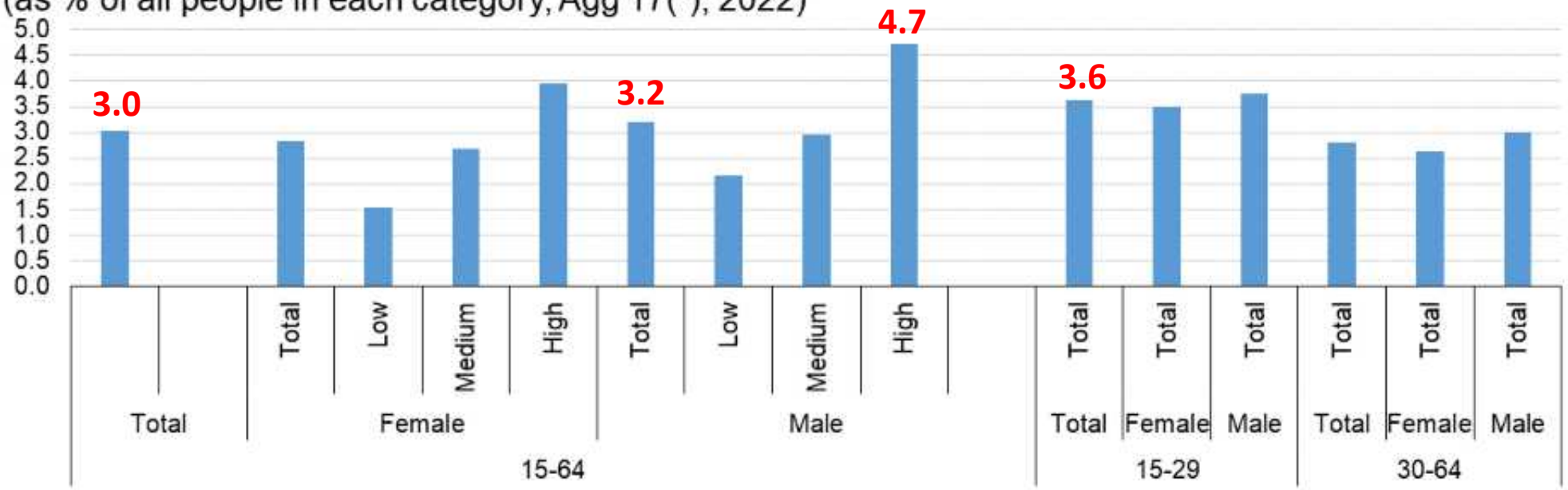


Note: (¹) Agg 17 refers to the aggregate of 17 countries listed in the data source section of the article, 'not stated' category is excluded
Source: LFS ad-hoc extraction - pilot survey

eurostat

Digital platform workers for at least 1h in the last year by age, sex and level of education

(as % of all people in each category, Agg 17(1), 2022)



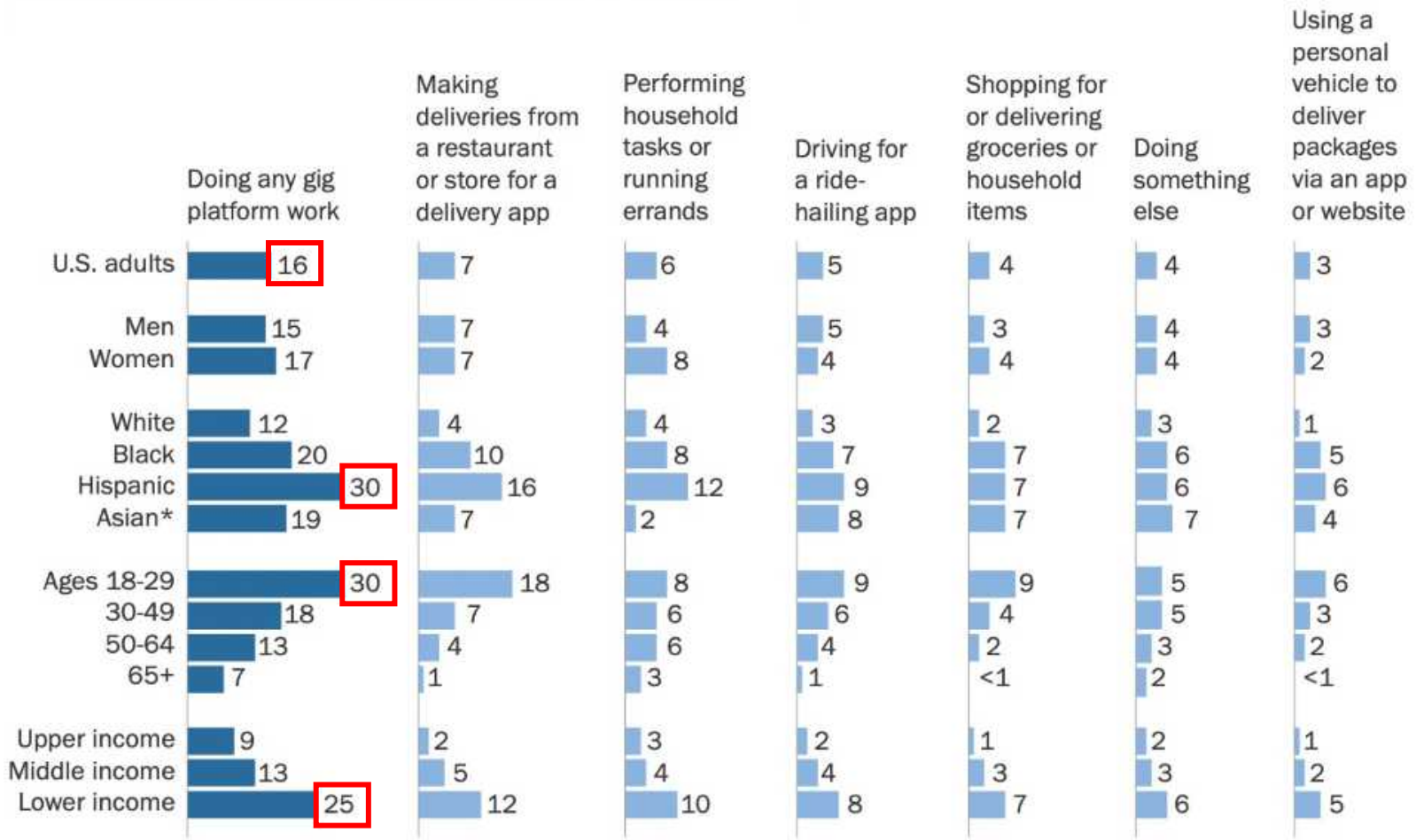
Note: (1) Agg 17 refers to the aggregate of 17 countries listed in the data source section of the article

Source: LFS ad-hoc extraction - pilot survey

eurostat

16% of Americans have ever earned money via an online gig platform; adults under 30, Hispanic adults and those with lower incomes especially likely to do these jobs

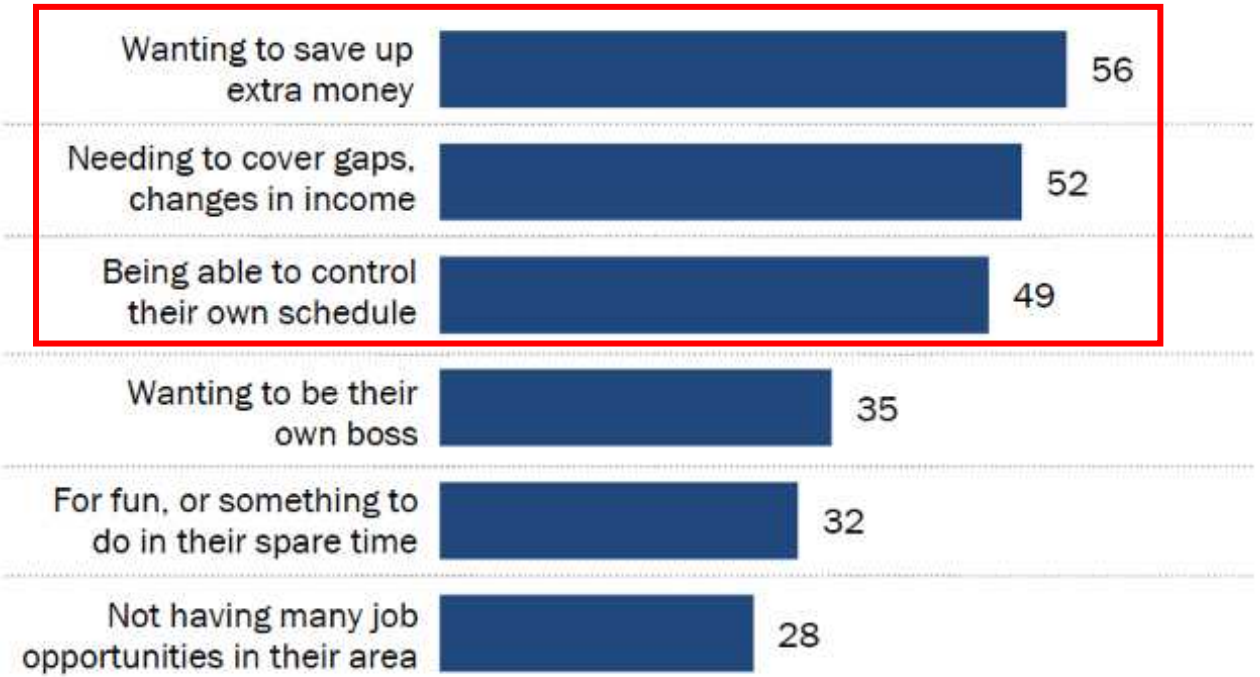
*% of U.S. adults who say they have **ever** earned money by ...*



A study surveyed 10,348 U.S. adults from Aug. 23 to 29, 2021 (Monica Anderson et al., 2021).

Saving up, covering gaps in income, flexible schedules top reasons current or recent gig platform workers cite for taking on these jobs over the past year

*Among gig platform workers who have earned money in the past 12 months on these online platforms, % who say each of the following is a **major reason** why they have taken on these jobs over the past 12 months*



According to statistics from the Occupational Safety and Health Administration, Ministry of Labor, Taiwan, the number of active delivery platform workers in Taiwan **grew rapidly** from around 45,000 in 2019 to 145,000 in 2022.

As of late 2025, the count reached approximately 150,000.



Journal of Epidemiology and Global Health (2025) 15:124
https://doi.org/10.1007/s44197-025-00468-w

RESEARCH

Assessing Carbon Monoxide Exposure in Food Delivery Workers by Using Exhaled Air with Consideration of Smoking Habits

Kun-Hua Li[1] · Ya-Yun Cheng[2] · Shih-Chieh Huang[1] · Chi-An Chen[1] · Jhih-Yuan Lu[1] · Zih-Ting Chao[1] · Yen-Cheng Tseng[3] · How-Ran Guo[1,4,5]

Although this study focused on CO exposure, the findings also revealed that many food delivery workers with **shorter tenures** had joined the **OFD** sector **during the COVID-19 pandemic**, indicating that their original employment or financial status was impacted by the social distancing policy and public concerns about the pandemic.

OPINION article

Front. Public Health, 09 February 2021

Sec. Occupational Health and Safety

Volume 9 - 2021 | https://doi.org/10.3389/fpubh.2021.638767

A Plea for the Need to Investigate the Health Effects of Gig-Economy

Anna Freni-Sterrantino *

Vincenzo Salerno

Department of Epidemiology and Biostatistics, Medical Research Council Centre for Environment and Health, Imperial College London, London, United Kingdom

*"Gig-workers represent a **vulnerable population** that is likely to be most exposed to **stress**."*

JAMA Cardiol
Published Online: January 5, 2022
2022;7;(2):125-126. doi:10.1001/jamacardio.2021.5435

Viewpoint

The Gig Economy Worker—A New Social Determinant of Health?

Fatima Rodriguez, MD, MPH[1]; Ashish Sarraju, MD[1,2]; Mintu P. Turakhia, MD, MAS[1,2,3,4]

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[3]Veterans Affairs Palo Alto Health Care System, Palo Alto, California

[4]Associate Editor, *JAMA Cardiology*

*"A man's **employment status** was a stronger predictor of his risk of dying from **coronary heart disease** than any of the more familiar risk factors."*

Determinants of Health

- Three key characteristics that may influence the health of platform workers
  - Work environment hazards
  - Business practices of digital labor platforms
  - Employment conditions

(Matilla-Santander et al., 2025)

Work Environment Hazards

- Psychosocial hazards
  - Social isolation, discrimination
- Ergonomic hazards
  - Heavy and repetitive work
- Physical hazards
  - Low or high temperatures
- Chemical hazards
  - Solvents, CO
- Digital tools related hazards
  - Technostress, screen exposure
- Traffic-related hazards
  - High-stress driving environments
- Atypical and long working hours
  - Circadian rhythm, sedentary periods
- Insufficient safety training
- Insufficient personal protective equipment

Business Practices of Digital Labor Platforms

- Algorithmic management
  - Intrusive surveillance
    - Feed algorithms
  - Algorithmic opacity
  - Rating systems
    - Remuneration
  - Loss of autonomy and control
  - Dehumanization
- Pricing strategy
  - Network effects
  - Gamification
- Inaccessible communication channels
  - Unilateral mechanisms

Employment Conditions

- Intermediary roles of digital labor platforms
  - Disguised self-employment
    - Experience close supervision
    - Receiving remuneration through the platforms
  - Limited access to social protection
    - Sickness benefits, occupational safety, and health protection
  - Low or unpredictable income
    - Bearing all job-related expenses
    - Multiple jobs

Social Science & Medicine

Date: November 2023
Article: 116281
Volume: Volume 337

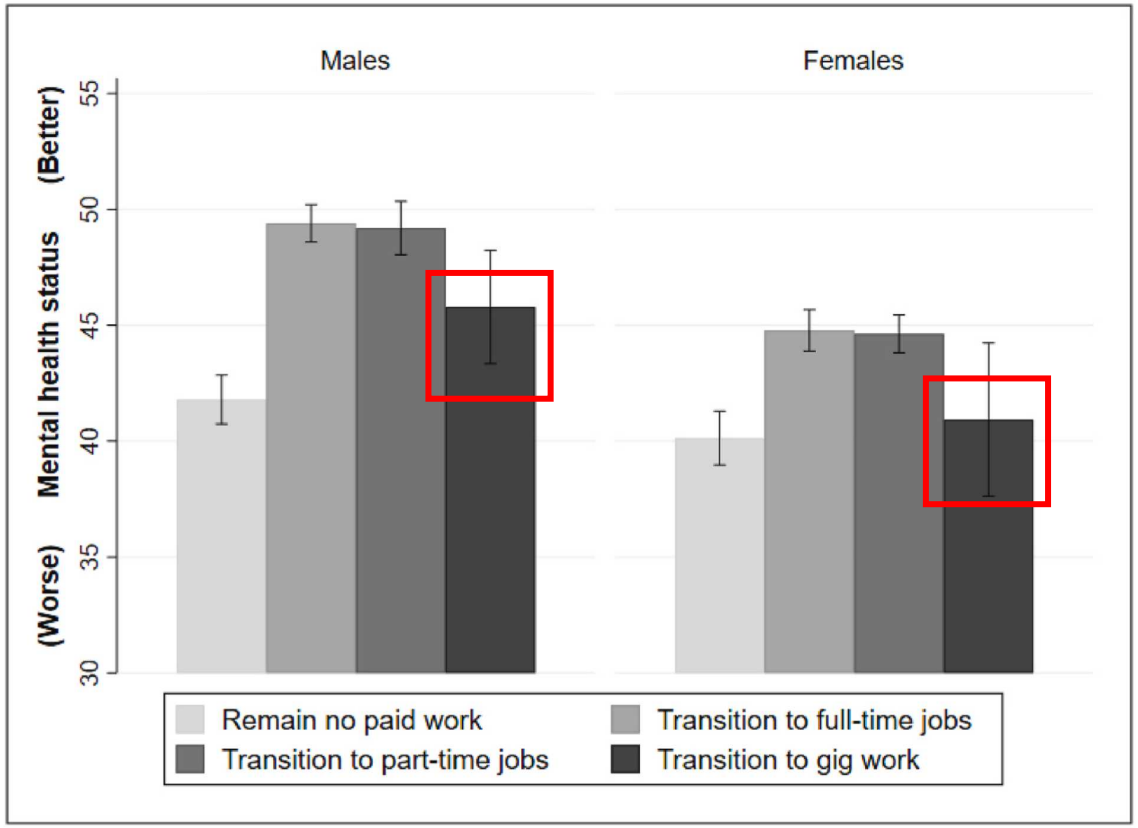
Research article ● Full text access Get rights and content ↗

Gig work and mental health during the Covid-19 pandemic: A gendered examination of comparisons with regular employment and unemployment

Zhuofei Lu [a] ✉, Senhu Wang [b] ✉, Wanying Ling [c d] ✉, Ya Guo [b] ✉

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[b] Department of Sociology and Anthropology, National University of Singapore, 11 Arts Link, #03-06 AS1, 117573, Singapore
[c] Department of Sociology, The University of Hong Kong, The Jockey Club Tower, Pokfulam Road, Hong Kong Special Administrative Region
[d] School of Sociology and Population Studies, Renmin University of China, No. 59 Zhongguancun Street, Haidian District, Beijing, 100872, PR China

- Gig work is related to **worse mental health** than transition into regular employment in **UK**.
- The findings revealed that **financial precarity** and **loneliness** can explain gig workers' **mental health decline**.



	Model 1
	Total
Panel A: Gig work with longer versus shorter working hours	
Work status transition (Ref. = Transition to gig work with shorter working hours ≤ 35 h)	
Remain no paid work	−2.53* (1.16)
Transition to full-time jobs	3.35** (1.14)
Transition to part-time jobs	3.07** (1.15)
Transition to gig work (>35 h)	−0.04 (2.79)
Constant	40.45*** (1.26)
Observations	3,381
R-squared	0.09

Safety Science

Date: February 2025
Article: 106722
Volume: Volume 182

Research article ● Full text access Get rights and content ↗

Psychosocial work conditions and health status of digital platform workers in Taiwan: A mixed method study

Yen-Ling Liu [a], Yawen Cheng [b], Ping-Hsiu Tsai [c], Yun-Chieh Yang [d], Ya-Chen Li [e], Wan-Ju Cheng [d f] Show more

- Platform workers exhibited **poorer** psychosocial work conditions, **burnout**, and **mental health statuses** than general workers in **Taiwan**.
- Challenges of platform workers include limited skill use, high **physical and psychological demands**, and **workplace violence**.

Table 3
Comparisons of work conditions and health in platform workers and general workers.

	Platform workers (N = 484)	General workers (N = 913)	*p*
	Mean (SD) /N (%)	*Mean (SD) /N (%)*	
Work conditions			
Job control, mean (SD)	44.57 (24.04)	55.60 (18.28)	<0.001
Skill discretion, mean (SD)	49.16 (21.89)	56.44 (19.05)	<0.001
Decision authority, mean (SD)	39.90 (30.26)	54.76 (22.17)	<0.001
Psychological job demands, mean (SD)	65.19 (17.36)	54.37 (20.30)	<0.001
High physical job demands	368 (76.03)	332 (36.36)	<0.001
Work rewards, mean (SD)	38.69 (22.19)	60.34 (15.53)	<0.001
Physical violence	12 (2.48)	11 (1.20)	0.119
Verbal violence	113 (23.40)	122 (13.36)	<0.001
Psychological violence	60 (12.40)	67 (7.34)	0.002
Sex harassment	24 (4.96)	13 (1.42)	<0.001
Health status			
Poor self-rated health	56 (11.57)	28 (3.07)	<0.001
Burnout score, mean (SD)	44.13 (21.72)	24.95 (20.59)	<0.001
Poor mental health (BSRS ≥ 6)	218 (45.04)	119 (13.03)	<0.001
Work-related injury	66 (13.64)	98 (10.73)	0.129

Ann Occup Environ Med. 2023 Oct 30;35:e43. doi: 10.35371/aoem.2023.35.e43

Association between exposure to violence, job stress and depressive symptoms among gig economy workers in Korea

Min-Seok Kim [1,2,3], Juyeon Oh [2], Juho Sim [2], Byung-Yoon Yun [4], Jin-Ha Yoon [3,4,✉]

Author information · Article notes · Copyright and License information

PMCID: PMC10654543 PMID: 38029274

Korean gig economic workers are exposed to **violence** and **job stress** in the workplace more than general workers, and the proportion of workers reporting **depressive symptoms** is also high.

Table 3. Univariate and multivariate logistic regression model of depressive symptoms

Variables	Odds ratio (95% confidence interval)	
	Model 1	Model 2
Workers type		
General	1.00 (reference)	1.00 (reference)
Gig	1.81 (1.25–2.62)	1.89 (1.19–3.00)
Sex		
Men		1.00 (reference)
Women		1.13 (0.74–1.73)
Age group (years)		
< 35		1.00 (reference)
< 45		1.56 (0.79–3.10)
< 55		2.03 (1.08–3.85)
< 65		1.69 (0.89–3.21
≥ 65		1.04 (0.46–2.39)
Working hours per week		
< 40		1.00 (reference)
< 52		0.84 (0.53–1.34)
≥ 52		1.20 (0.75–1.92)
Education		
Highschool or below		1.00 (reference)
College		1.30 (0.78–2.19)
University		1.42 (0.88–2.28)
Graduate school		0.85 (0.31–2.37)

Table 4. Multivariable logistic regression model of depressive symptoms among gig workers

Variables	Odds ratio[a] (95% confidence interval)
Verbal violence	
0	1.00 (reference)
≥ 1	1.37 (0.67–2.81)
Humiliating treatment	
0	1.00 (reference)
≥ 1	3.53 (1.67–7.45)
Blasphemy experience	
0	1.00 (reference)
≥ 1	0.68 (0.32–1.44)
Physical violence	
0	1.00 (reference)
≥ 1	1.22 (0.53–2.81)
Sexual harassment	
0	1.00 (reference)
≥ 1	2.65 (1.19–5.88)
Forced to work beyond contract	
Low frequency	1.00 (reference)
High frequency	1.22 (0.65–2.31)
Meals per day	
≥ 3	1.00 (reference)
< 3	3.55 (2.02–6.27)

Safety and Health at Work

Date: March 2024
Pages: 17-23
Volume: Volume 15, Issue 1

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Original article

Investigation of Working Conditions and Health Status in Platform Workers in the Republic of Korea

Hyoungseob Yoo [1], Munyoung Yang [1], Ji-Hun Song [2], Jin-Ha Yoon [3], Wanhyung Lee [4], Jinhee Jang [5], Minjoo Yoon [6], Mo-Yeol Kang [1] Show more

- Most **Korean** platform workers existed in the blind spot of social insurance, facing frequent exposure to **physical and chemical hazards**, **ergonomic risk factors**, and direct or indirect **violence**.
- The prevalence of health problems, including **musculoskeletal symptoms**, **general fatigue**, and **depressive symptoms**, in each occupational group was statistically higher than that in the general population.

Table 4
Standardized prevalence ratio (95% CI) of the reported symptoms by the three types of platform workers*

Symptoms	Designated drivers	Food delivery drivers	Housekeeping managers
Back pain	0.68 (0.57–0.80)	**1.40 (1.16–1.67)**	**1.26 (1.00–1.56)**
Upper limb pain	**1.35 (1.15–1.57)**	**1.67 (1.42–1.94)**	**2.78 (2.30–3.33)**
Lower limb pain	**1.71 (1.40–2.06)**	**2.42 (2.00–2.91)**	**5.26 (4.04–6.73)**
Headache, eye fatigue	**2.23 (1.91–2.59)**	**1.54 (1.27–1.85)**	0.86 (0.58–1.22)
Anxiety	**1.84 (1.34–2.46)**	1.29 (0.86–1.86)	0.67 (0.27–1.39)
Fatigue	**1.69 (1.45–1.96)**	**2.12 (1.82–2.45)**	**1.68 (1.35–2.05)**
Depressive symptoms	**4.19 (3.24–5.33)**	**3.10 (2.40–3.94)**	**1.85 (1.01–3.10)**

Bold fonts indicate statistically significant results.

* Depressive symptoms (PHQ-9>5) were compared to the results of the Republic of Korea National Health and Nutrition Examination Survey. Other symptoms were compared with the results of the 6th Korean Working Conditions Survey (2020).

J Occup Environ Med. 2020 May;62(5):e236-e239. doi: 10.1097/JOM.0000000000001849.

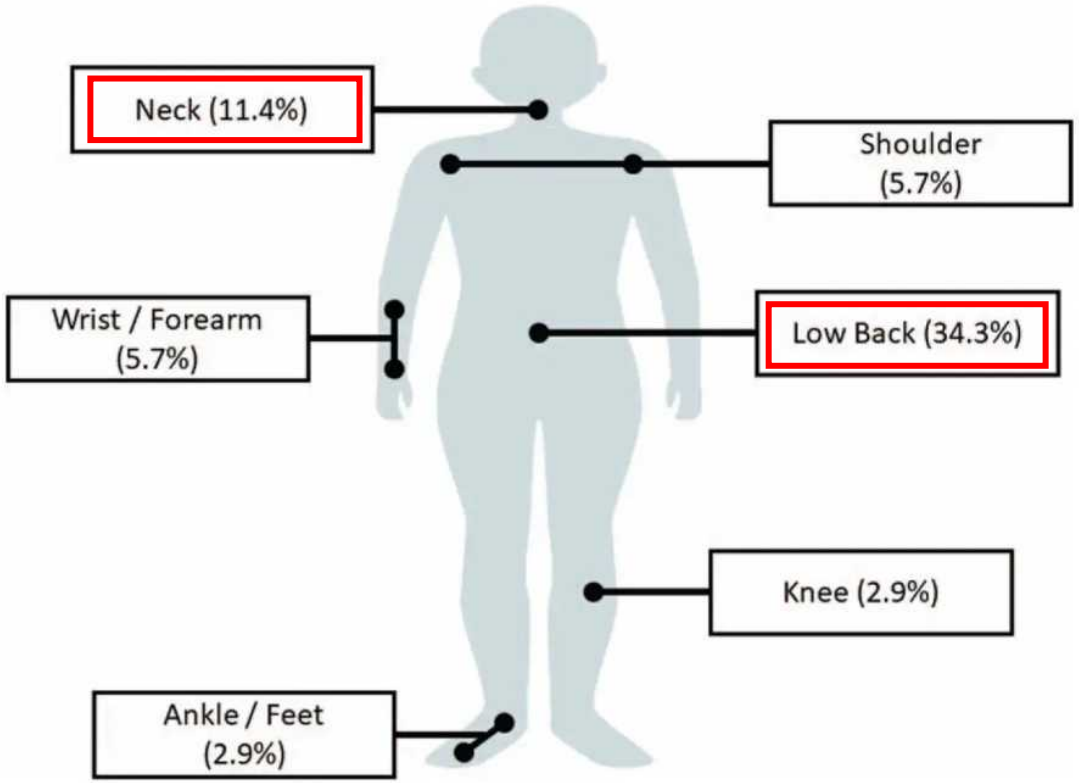
Acute Musculoskeletal Pain Reported Among Rideshare Drivers in the Health/Safety Investigation Among Non-Standard Workers in the Gig Economy (H.I.N.G.E.) Pilot Study

Alberto J Caban-Martinez, Katerina M Santiago, Paola Louzado Feliciano, Kemi Ogunsina, Hannah Kling, Kevin Griffin, Natasha Schaefer Solle

PMID: 32149940 DOI: 10.1097/JOM.0000000000001849

This study found approximately a third of rideshare drivers surveyed in this **US** pilot sample had experienced **acute musculoskeletal pain** (in the prior 7-days to survey administration) of which **low back** and **neck** were the most frequently reported anatomic sites.

Characteristics	Total Sample *N* (%)*	Yes, Acute Joint Pain *n* (%)*	No Acute Joint Pain *n* (%)*	*P*-Value
Row total	35 (100.0)	13 (37.1)	22 (62.9)	
Age				0.849
21–34 years old	7 (20.0)	3 (23.1)	4 (18.2)	
35–54 years old	21 (60.0)	7 (53.8)	14 (63.6)	
55 and older	7 (20.0)	3 (23.1)	4 (18.2)	
Gender				0.392
Male	27 (77.1)	9 (69.2)	18 (81.8)	
Female	8 (22.9)	4 (30.8)	4 (18.2)	
Self-rated general health status				0.008
Excellent	13 (37.1)	1 (7.7)	12 (54.5)	
Very good/good	20 (57.1)	10 (76.9)	10 (45.5)	
Fair/poor	2 (5.7)	2 (15.4)	0 (0.0)	
Job/Work characteristics				
Rideshare primary job				0.022
Yes	21 (60.0)	11 (84.6)	10 (45.5)	
No	14 (40.0)	2 (15.4)	12 (54.5)	
Work part-/full-time				0.267
Part-time	15 (42.9)	4 (30.8)	11 (50.0)	
Full-time	20 (57.1)	9 (69.2)	11 (50.0)	
Have a second job				0.631
Yes	18 (51.4)	6 (46.2)	12 (54.5)	
No	17 (48.6)	7 (53.8)	10 (45.5)	
Takes breaks outside car				0.349
Always	2 (5.7)	1 (7.7)	1 (4.5)	
Occasionally	24 (68.6)	7 (53.8)	17 (77.3)	
Rarely	9 (25.7)	5 (38.5)	4 (18.2)	



> Int J Occup Saf Ergon. 2024 Sep;30(3):825-834. doi: 10.1080/10803548.2024.2356997. Epub 2024 Jun 10.

The hidden cost of your 'too fast food': stress-related factors and fatigue predict food delivery riders' occupational crashes

Sergio A Useche [1 2], Sebastián Robayo [3], Mauricio Orozco-Fontalvo [4]

Affiliations – collapse

Affiliations

1 Faculty of Psychology, University of Valencia, Valencia, Spain.
2 Research Institute on Traffic and Road Safety (INTRAS), University of Valencia, Valencia, Spain.
3 Colombian Ministry of Sports, Bogotá, Colombia.
4 Higher Technical Institute, University of Lisbon, Lisbon, Portugal.

The findings showed that the **occupational (riding) crashes** of food delivery riders in **Spain** can be largely explained through **work-related fatigue**, which exerts a full mediation between job settings, **stress-related factors** and riding safety outcomes.

Table 2. Bivariate (Spearman's ρ) correlations among psychosocial work-related study variables.

Study variable		Mean (*SD*)	1	2	3	4	5	6
1	Job tenure (years)	1.94 (1.05)	–	–	–	–	–	–
2	Urban insecurity incidents	1.46 (0.79)	0.122	–	–	–	–	–
3	Effort–reward imbalance	1.07 (0.25)	−0.060	−0.114	–	–	–	–
4	Job strain	1.37 (0.28)	0.024	−0.025	0.112	–	–	–
5	Psychological distress	16.78 (3.75)	−0.034	0.121	0.142*	0.163*	–	–
6	Need for recovery	3.81 (2.47)	0.146*	0.141*	0.222**	0.098	0.190**	–
7	Work traffic crashes	0.35 (0.78)	0.084	0.227**	0.167**	0.091	0.182**	0.205**

ELSEVIER

Journal of Transport & Health
Volume 13, June 2019, Pages 115-127

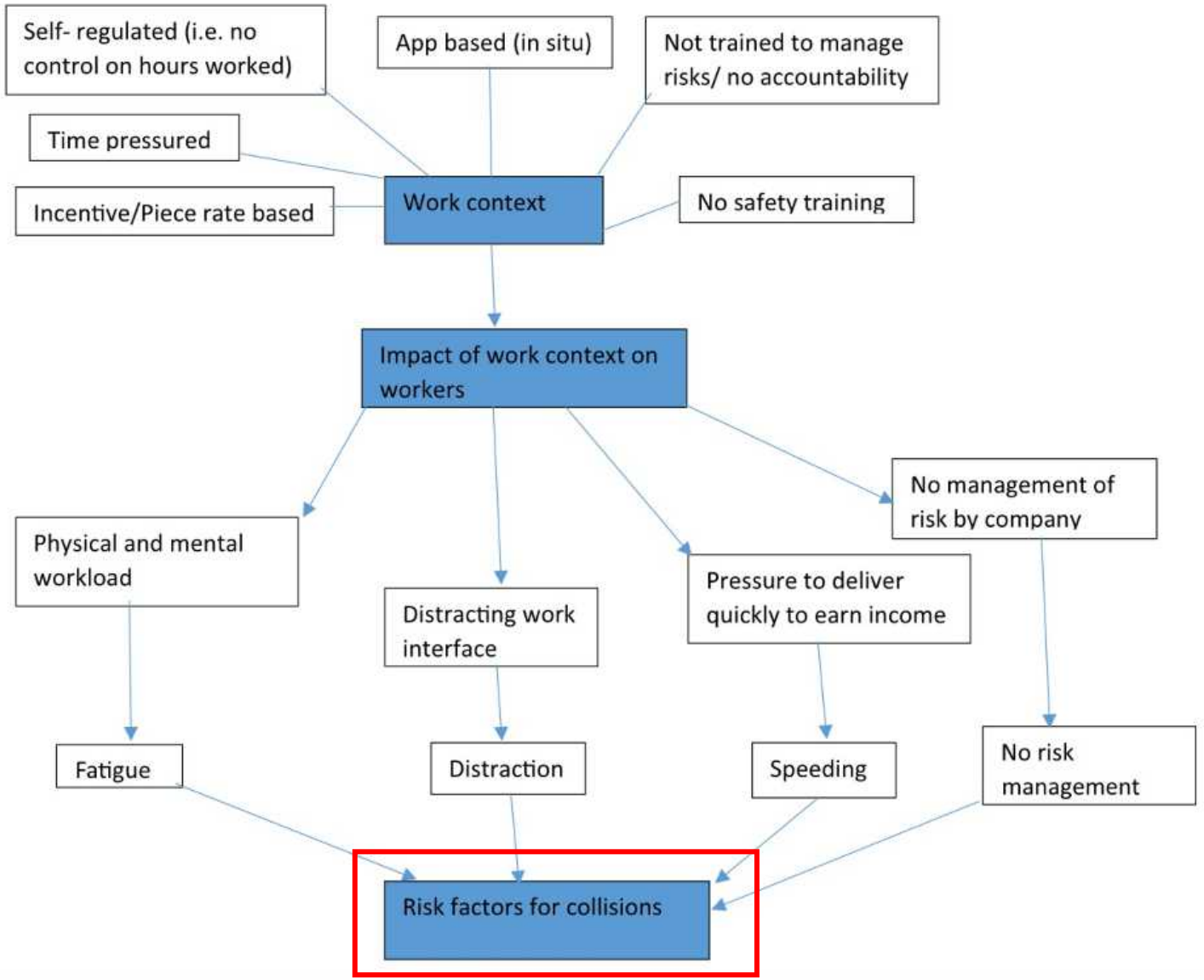


The health and safety risks for people who drive for work in the gig economy

Nicola Christie, Heather Ward

Centre for Transport Studies, Dept. Civil, Environmental & Geomatic Engineering, University College London, Chadwick Building, Room 209, Gower St, London WC1E 6BT, UK

- The results showed that **UK** gig economy drivers experience **fatigue**, **pressure** to violate traffic regulations and **distraction** from their phones.
- Young people who work using two-wheeled vehicles seem particularly vulnerable to **collisions**.


Self- regulated (i.e. no control on hours worked)
App based (in situ)
Not trained to manage risks/ no accountability
Time pressured
Incentive/Piece rate based
Work context
No safety training
Impact of work context on workers
Physical and mental workload
Distracting work interface
Pressure to deliver quickly to earn income
No management of risk by company
Fatigue
Distraction
Speeding
No risk management
Risk factors for collisions

Safety Science

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Volume: Volume 158

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Delivering hot food on motorcycles: A mixed method study of the impact of business model on rider behaviour and safety

Nicola Christie, Heather Ward

Centre for Transport Studies, Chadwick 209, UCL, Gower street, London WC1E 6BT, United Kingdom

- Motorcyclists delivering food report via apps feel **pressurized** and take risks in **US** samples.
- The app causes a high level of **distraction** for motorcyclists delivering food.
- Motorcyclists delivering food via apps report that they **violate traffic laws**.
- Employed riders feel less pressure, violate less and report less **collisions**.

Table 3
Pressure to speed model.

Explanatory variables	B	S.E.	Wald	Df	Sig.	Exp(B)	95 % C.I. for EXP(B)	
							Lower	Upper
Cc			2.148	3	0.542			
cc(1)	-0.500	0.764	0.428	1	0.513	0.607	0.136	2.712
cc(2)	-0.732	0.732	0.998	1	0.318	0.481	0.115	2.021
cc(3)	−1.063	0.824	1.661	1	0.197	0.346	0.069	1.739
molic(1)	-0.378	0.310	1.487	1	0.223	0.685	0.373	1.258
longlic			2.205	2	0.332			
longlic(1)	-0.548	0.374	2.139	1	0.144	0.578	0.278	1.205
longlic(2)	-0.323	0.319	1.030	1	0.310	0.724	0.388	1.351
weekh			2.127	3	0.546			
weekh(1)	0.823	0.784	1.101	1	0.294	2.277	0.490	10.592
weekh(2)	0.618	0.736	0.704	1	0.401	1.855	0.438	7.855
weekh(3)	0.933	0.760	1.509	1	0.219	2.543	0.574	11.277
Ethnic (1)	-0.155	0.287	0.292	1	0.589	0.857	0.488	1.502
Sex (1)	-0.374	0.344	1.183	1	0.277	0.688	0.350	1.350
Age	-0.043	0.017	6.222	1	**0.013**	0.958	0.926	0.991
Comp(1)	0.652	0.260	6.265	1	**0.012**	1.919	1.152	3.198
Chance			37.577	2	0.000			
Chance (1)	1.985	0.328	36.666	1	**0.000**	7.276	3.827	13.830
Chance (2)	0.469	0.306	2.352	1	0.125	1.598	0.878	2.910
Constant	1.028	1.212	0.720	1	0.396	2.797		

Comp	Type of company	1 = gig, 2 = employed
Chance	Agreement level that they have a high chance of being stopped by the police while working	1 = agree, 2 = neither agree or disagree, 3 = disagree

J Urban Health. 2024 Apr 29;101(3):439–450. doi: 10.1007/s11524-024-00873-9

A Price Too High: Injury and Assault among Delivery Gig Workers in New York City

Zoey Laskaris [1,✉], Mustafa Hussein [2], Jim P Stimpson [3], Emilia F Vignola [4], Zach Shahn [5], Nevin Cohen [2], Sherry Baron [1]

- About 21.9% and 20.8% of **US** respondents reported **being injured** and **assaulted**, respectively.
- Fully dependent food delivery gig workers, especially two-wheeled riders, are highly vulnerable to the negative consequences of working conditions under algorithmic management by the platforms.

	Injury[a]		
	Model 1[b]	Model 2[b]	Model 3[b]
	aPR [95% CI]	aPR [95% CI]	aPR [95% CI]
Dependence			
Side job	1 [1.00, 1.00]	1 [1.00, 1.00]	1 [1.00, 1.00]
Main job	2.39*** [1.83, 3.12]	2.02*** [1.54, 2.64]	1.61** [1.20, 2.16]
Age (years)			
18–24	1 [1.00, 1.00]	1 [1.00, 1.00]	1 [1.00, 1.00]
25–34	0.85 [0.67, 1.09]	0.91 [0.72, 1.16]	0.87 [0.69, 1.11]
35–44	0.76 [0.58, 1.01]	0.85 [0.65, 1.12]	0.82 [0.63, 1.07]
45 and older	0.75 [0.54, 1.04]	0.97 [0.71, 1.33]	0.93 [0.68, 1.28]
Mode of transport			
Car		1 [1.00, 1.00]	1 [1.00, 1.00]
Moped, e-bike		2.34*** [1.90, 2.88]	2.32*** [1.89, 2.85]
Usual weekly hours			
Less than 20 h			1 [1.00, 1.00]
20–39 h			1.23 [0.94, 1.62]
40 or more hours			1.81*** [1.39, 2.36]

Ind Health. 2022 May 11;60(4):360–370. doi: 10.2486/indhealth.2022-0012

Relationship between occupational injury and gig work experience in Japanese workers during the COVID-19 pandemic: a cross-sectional internet survey

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- The findings revealed that **Japanese** workers involved in gig work had a greater risk of any minor **occupational injuries** and **activity-limiting injuries** than those not involved in gig work, after adjusting for covariates.
- The results of this study indicate that gig workers are exposed to greater **occupational hazards** during the COVID-19 pandemic.

Table 3. Relationship between gig work experience and occupational injury

		Any minor injury			Activity-limiting injury		
		OR	95% CI		OR	95% CI	
Gig work experience in the last year	No	1.00	(reference)		1.00	(reference)	
	Yes	3.68	3.02	4.49	9.11	7.03	11.8
Age (year)	–19	2.17	1.54	3.07	1.48	0.70	3.09
	20–29	1.85	1.58	2.16	2.17	1.64	2.89
	30–39	1.00	(reference)		1.00	(reference)	
	40–49	0.85	0.72	0.99	0.68	0.50	0.94
	50–59	0.69	0.58	0.82	0.66	0.47	0.93
	60–69	0.79	0.65	0.97	0.89	0.61	1.29
	70–79	0.72	0.54	0.97	0.51	0.27	0.94
Sex	Female	1.00	(reference)		1.00	(reference)	
	Male	0.97	0.86	1.10	0.92	0.73	1.16
Household income in 2020 (1,000yen)	6,000+	1.00	(reference)		1.00	(reference)	
	3,000–5,999	1.38	1.21	1.57	1.02	0.80	1.30
	–2,999	1.52	1.29	1.79	1.14	0.84	1.54
	Do not want to answer	1.04	0.85	1.28	0.67	0.44	1.02
	Do not know	1.23	1.02	1.47	0.87	0.61	1.24

Work, Industrial & Organizational Psychology

Navigating the gig economy: exploring challenges and motivations for the wellbeing of Gen Y and Gen Z gig workers

Jikku Susan Kurian & N. Bindu Madhavi

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- Motivation positively impacts quality of life and has a positive effect on well-being of younger gig workers in **India**.
- The findings indicated that **work-related challenges (e.g., workload)** have a negative influence on **stress** and a subsequent **negative effect on well-being**.

	Wellbeing	Motivation	Challenges	Quality of life	Stress	Gig workers attributes
Wellbeing	**0.842**					
Motivation	0.448**	**0.894**				
Challenges	−0.470**	−0.456**	**0.789**			
Quality of life	0.182**	0.289**	−0.203**	**0.789**		
Stress	−0.502**	−0.230**	0.242**	−0.052	**0.777**	
Gig workers attributes	−0.362**	−0.433**	0.379**	−0.260**	0.486**	**0.900**

> Front Psychol. 2021 Dec 2:12:724966. doi: 10.3389/fpsyg.2021.724966. eCollection 2021.

Flexible, Self-Determined... and Unhealthy? An Empirical Study on Somatic Health Among Crowdworkers

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Affiliations – collapse

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Crowd workers show significantly **increased somatic symptoms** as compared to a **German** norm sample, that are stable across different kinds of tasks and platforms, gender, and age groups, and that is statistically due to **the extent of participation** in crowd work.

TABLE 4 | Hierarchical regression results for somatic symptoms.

Variable	*B*	95% CI for B		SE B	*β*	R^2	ΔR^2
		LL	UL				
Step 1						0.044	0.04***
Constant	10.83	9.02	12.64	0.92			
Gender	−2.17	−2.91	−1.42	0.38	−0.21***		
Age	−0.04	−0.07	−0.01	0.02	−0.09*		
Step 2						0.056	0.01*
Constant	10.82	9.01	12.63	0.92			
Gender	−1.93	−2.69	−1.16	0.39	−0.18***		
Age	−0.04	−0.08	−0.01	0.02	−0.10*		
Hours CW	0.06	0.02	0.09	0.02	0.11**		
Total work hours	−0.02	−0.04	0.00	0.01	−0.07		
Step 3						0.163	0.11***
Constant	5.85	3.84	7.85	1.02			
Gender	−1.65	−2.37	−0.93	0.37	−0.16***		
Age	−0.02	−0.05	−0.01	0.02	−0.05		
Hours CW	0.07	0.04	0.11	0.02	0.15***		
Total work hours	−0.04	−0.05	−0.02	0.01	−0.14***		
WFC-time	−0.00	−0.46	0.45	0.23	0.00		
WFC-strain	1.75	1.27	2.24	0.25	0.34***		

CI, confidence interval; LL, lower limit; UL, upper limit; Hours CW, hours doing crowdwork per week; WLC-time/−strain, time-based and strain-based work-life conflict.
$^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$.

Current Evidence on Adverse Health Effects

- Work Environment Hazards
  - Compared with transitioning to regular employment, platform work is associated with **poorer mental health** in UK, primarily driven by financial precarity and loneliness (Lu et al., 2023).
  - Compared to general workers, Taiwanese platform workers report **poorer mental health** and heightened **burnout**, fueled by intense physical and psychological demands, and workplace violence (Liu et al., 2025).
  - Korean platform workers faced greater workplace violence, job stress, and **depressive symptoms** than traditional workers (Kim et al., 2023).

Current Evidence on Adverse Health Effects

- Work Environment Hazards (Cont.)
  - Korean platform workers face heightened occupational hazards—including ergonomic strain, violence, and environmental risks—and suffer significantly higher rates of **musculoskeletal symptoms**, **fatigue**, and **depression** than the general population (Yoo et al., 2024).
  - In a US pilot sample, roughly one-third of rideshare drivers reported past-week **musculoskeletal pain**, primarily in the **lower back** and **neck** (Caban-Martinez et al., 2020).
  - In a Spain sample, work-related fatigue fully mediates the impact of job characteristics and stress on food delivery riders' **crash** risks (Useche et al., 2024).

Current Evidence on Adverse Health Effects

- Work Environment Hazards (Cont.)
  - UK gig drivers contend with fatigue, phone distractions, and pressure to break traffic laws; young two-wheeled couriers face the highest **collision** risk (Christie et al., 2019).
  - App-driven pressure and distraction push US food delivery workers toward risk-taking and traffic violations, significantly driving up **collision** rates (Christie et al., 2023).
  - Subject to work condition, fully dependent two-wheeled food delivery couriers in US face severe vulnerabilities, with 21.9% reporting **injuries** and 20.8% reporting physical **assaults** (Laskaris et al., 2024).

Current Evidence on Adverse Health Effects

- Work Environment Hazards (Cont.)
  - Adjusted analyses show Japanese gig workers faced higher risks of minor and **activity-limiting occupational injuries** than non-gig workers, underscoring heightened workplace hazards during COVID-19 (Morita et al., 2022).
  - Motivation improves Indian platform workers' quality of life and well-being, while work-related challenges exacerbate **stress**, ultimately **eroding well-being** (Kurian et al., 2024).
  - Compared to German norms, platform workers exhibit significantly **elevated somatic symptoms** across all demographics and task types, with symptom severity driven by the extent of their platform involvement (Schlicher et al., 2021).

New Technology, Work and Employment

RESEARCH ARTICLE OPEN ACCESS

Algorithmic Paranoia: Gig Workers' Affective Experience of Abusive Algorithmic Management

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- Gig workers increasingly report feelings of mistrust, anxiety, and profound fear under opaque and abusive **algorithmic management**.
- Platform workers reported **poor mental well-being** due to exposure to algorithmic management, including the automation of client-matching processes, **rating systems**, and **unilateral dispute resolution mechanisms**.

> Int J Qual Stud Health Well-being. 2023 Dec;18(1):2173336.
doi: 10.1080/17482631.2023.2173336.

Psychosocial work environment and mental wellbeing of food delivery platform workers in Helsinki, Finland: A qualitative study

Benta Mbare [1]

Affiliations + expand

PMID: 36730307 PMCID: PMC9897739 DOI: 10.1080/17482631.2023.2173336

- A psychosocially burdening work environment is detrimental to the mental wellbeing of food delivery platform workers.
- Experiencing **the lack of human contact** with platform and working in an onerous work environment accentuated the **stress** and **poor mental health** among platform workers.

Research Article

Algorithmic management, wellbeing and platform work: understanding the psychosocial risks and experiences of food couriers in Finland

Benta Mbare, Mikko Perkiö & Meri Koivusalo

Pages 386-411 | Received 30 Jun 2024, Accepted 27 Sep 2024, Published online: 07 Nov 2024

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- The findings showed that **algorithmic management** for monitoring and evaluation purposes increased the work pace and work demands and generated **psychological** and **physical job strains**.
- Algorithmic management had direct and indirect intertwined negative psychosocial influence on food couriers, causing **poor mental health** and **physical strain**.

ELSEVIER

Journal of Transport & Health
Volume 14, September 2019, 100571



Stressful by design: Exploring health risks of ride-share work

Emma Bartel [a], Ellen MacEachen [a], Emily Reid-Musson [a], Samantha B. Meyer [a], Ron Saunders [b], Philip Bigelow [a], Agnieszka Kosny [b c], Sharanya Varatharajan [a]

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[c] School of Public Health, University of Toronto, Toronto, Canada

- The findings revealed that **gamification** incentivized drivers to work longer hours, and they associated this with **sedentary behavior**.
- Health risks for ride-share drivers included **stress**, **fatigue**, **musculoskeletal disorders**, and **urinary disorders**.

ORIGINAL ARTICLE

Digital industrial accidents: A case study of the mental distress of platform workers in South Korea

Suyoung Kim ✉, Myungjoo Kang, Joonhyeog Park

First published: 08 December 2021 | https://doi.org/10.1111/ijsw.12522 | VIEW METRICS

A study investigated platform workers' **mental distress** and identified five common psychological problems: (a) **stress** from real-time assessment in public space; (b) **exhaustion** from working without work-life boundaries; (c) **infringement of privacy** through exposure of private life; (d) **verbal abuse** in non-face-to-face relations; and (e) **isolation** from working alone.

Current Evidence on Adverse Health Effects

- Business Practices of Digital Labor Platforms
  - Opaque algorithmic controls—including automated client matching, rating systems, and unilateral dispute mechanisms—foster intense **anxiety**, **fear**, and **mistrust** among gig workers, severely undermining their **mental health** (Alacovska et al., 2025).
  - Onerous working conditions and isolation from platform support intensify **stress**, severely impairing food delivery workers' **mental health** (Mbare, 2023).
  - Algorithmic tracking and evaluation intensify work demands and pace, exerting compounded psychosocial pressures that drive **physical strain** and **mental distress** among food couriers (Mbare et al., 2024).

Current Evidence on Adverse Health Effects

- Business Practices of Digital Labor Platforms (Cont.)
  - Platform gamification drives rideshare operators toward prolonged sedentary hours, heightening risks of **stress**, **fatigue**, **musculoskeletal disorders**, and **urinary conditions** (Bartel et al., 2019).
  - Platform workers' **mental distress** stems from five core stressors: real-time public evaluation, work-life boundaryless exhaustion, privacy infringement, remote verbal abuse, and isolation (Kim et al., 2021).

> Med Lav. 2024 Oct 24;115(5):e2024035. doi: 10.23749/mdl.v115i5.16278.

Occupational Safety and Health of Riders Working for Digital Food Delivery Platforms in the City of Milan, Italy

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4 EPIGET Lab, Department of Clinical Sciences and Community Health - University of Milan, Milan, Italy/Occupational Health Unit - Fondazione IRCCS Ca' Granda Ospedale Maggiore Policlinico, Milan, Italy.

- Road **accidents** were reported by 39% of riders, influenced by type of vehicle, fatigue, and number of daily deliveries, and **employment conditions**.
- Physical and verbal **assaults** (12% and 28%, respectively) were reported, as well as health-related issues, particularly **musculoskeletal disorders**.

Safety Science

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Research article Get rights and content ↗

More reliance, more injuries: Income dependence, workload and work injury of online food-delivery platform riders

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[b] Qu Qiubai School of Government, Changzhou University, China

- Food-delivery riders' **income dependence degree** is positive related to **work injury**.
- The results indicated that **workload** mediates the relationship between income dependence and work injury.

Current Evidence on Adverse Health Effects

- Employment Conditions
  - Riders frequently faced **road accidents** (39%, influenced by fatigue, vehicle type, delivery volume, and working conditions), **verbal abuse** (28%), **physical assaults** (12%), and **musculoskeletal disorders** (Boniardi et al., 2024).
  - Food delivery riders' degree of income dependence is positively associated with **occupational injuries**, a relationship fully mediated by workload (Jing et al., 2023).

Limitations of Current Evidence

- Most studies focused on case reports and qualitative interviews
- A cross-sectional design was used in most quantitative studies
- Most quantitative studies did not include a reference group
- Lack of validated measurement instruments
- Inadequate sample sizes

Potential Adverse Health Effects

- Short-term effects
  - Symptoms of depression, stress, fear, and anxiety
  - Sleep disorders
  - Occupational injuries
  - Musculoskeletal pain
  - Poor general well-being with somatic symptoms
- Medium-term effects
  - Cardiometabolic risks
    - Dyslipidemia, hypertension, atherosclerosis

Potential Adverse Health Effects

- Long-term effects
  - Mental disorders
  - Cardiovascular diseases
  - Cerebrovascular diseases
  - Metabolic disorders
  - Musculoskeletal disorders
  - Urinary disorders

Conclusions

- Literature shows a link between digital platform work and poor health
- The current evidence, mainly focused on mental health and location-based platform workers, highlights factors contributing to poor physical and mental health
  - Low-quality employment conditions, opaque algorithmic management, and psychosocial work environment hazards
- Large-scale quantitative and longitudinal studies with a comparison group are needed

Thank you for your attention.